GCSE Mathematics - Foundation Tier

Paper 2 (Calculator) Practice - Set 4

Time: 1 hour 30 minutes | Total marks: 80 | Calculator allowed

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Instructions

- Use black ink or ball-point pen.
- Answer ALL questions in the spaces provided.
- You must show all your working - answers given without working may not score full marks.
- Diagrams are NOT accurately drawn unless otherwise stated.
- If your calculator does not have a pi button, use pi = 3.142.

1.

Write 0.45 as a percentage.

(Total for Question 1 is 1 marks)

2.

Work out 12.5% of £640.

(Total for Question 2 is 2 marks)

3.

(a) Round 4.836 to 1 significant figure. (1)

(b) Round 0.0719 to 2 significant figures. (1)

(Total for Question 3 is 2 marks)

4.

Find the lowest common multiple (LCM) of 6 and 8.

(Total for Question 4 is 2 marks)

5.

The frequency table shows the number of goals scored by a football team in 20 matches.

Goals	0	1	2	3	4
Freq.	3	7	6	3	1

(a) Write down the modal number of goals. (1)

(b) Work out the total number of goals scored across the 20 matches. (2)

(Total for Question 5 is 3 marks)

6.

Expand $3(2x - 5)$.

(Total for Question 6 is 2 marks)

7.

A cyclist rides a distance of 30 km in 1 hour 20 minutes.

Work out the average speed of the cyclist in km/h.

(Total for Question 7 is 3 marks)

8.

The ages (in years) of 10 customers at a coffee shop are:

18 22 25 27 30 33 34 38 45 52

(a) Find the median age. (2)

(b) Find the range. (1)

(Total for Question 8 is 3 marks)

9.

Lara, Mia and Noah share £288 in the ratio 5 : 3 : 4.

Work out the amount that Mia receives.

(Total for Question 9 is 3 marks)

10.

Solve $3x + 4 = 2x + 17$.

You must show all your working.

(Total for Question 10 is 3 marks)

11.

A trapezium has parallel sides of length 7 cm and 11 cm, and a perpendicular height of 6 cm.

[Diagram: trapezium with parallel sides 7 cm (top) and 11 cm (bottom), perpendicular height 6 cm.]

Work out the area of the trapezium.

(Total for Question 11 is 3 marks)

12.

A pie chart shows how 180 people travel into a city centre.

[Diagram: pie chart with sectors - Train: 144 deg, Bus: 108 deg, Cycle: 72 deg, Other: 36 deg.]

(a) How many people travel by Train? (2)

(b) What fraction of people Cycle? Give your answer in its simplest form. (2)

(Total for Question 12 is 4 marks)

13.

The formula $v = u + at$ is used in motion problems.

(a) Work out v when $u = 8$, $a = 3$ and $t = 4$. (2)

The formula $C = 5(F - 32) / 9$ converts degrees Fahrenheit (F) into degrees Celsius (C).

(b) Work out C when $F = 86$. (2)

(Total for Question 13 is 4 marks)

14.

The population of a village in 2020 was 4500. By 2026 the population had risen to 5085.

(a) Work out the percentage increase in the population. Give your answer to 1 decimal place. (3)

A neighbouring village had a population of 3800 in 2020. Its population increased by 8% by 2026.

(b) Work out the population of the neighbouring village in 2026. (2)

(Total for Question 14 is 5 marks)

15.

Here are the first four terms of an arithmetic sequence.

7, 11, 15, 19, ...

(a) Write down the next term. (1)

(b) Find an expression for the nth term of the sequence. (2)

(c) Work out the 50th term of the sequence. (2)

(Total for Question 15 is 5 marks)

16.

A recipe for 8 portions of pasta sauce uses 600 g of tomatoes.

(a) Work out the mass of tomatoes needed for 14 portions. (2)

A jar of tomato paste replaces 75 g of fresh tomatoes.

(b) Work out how many jars are needed to replace ALL the fresh tomatoes in a 14-portion recipe. (2)

(Total for Question 16 is 4 marks)

17.

The table shows the outside temperature (deg C) and the daily ice-cream sales (£) at a beach kiosk on 8 days.

Temp.	16	18	20	22	24	26	28	30
Sales	120	155	180	220	260	295	330	370

(a) On the grid provided, plot these data as a scatter graph. (2)

(b) Describe the correlation between temperature and ice-cream sales. (1)

(c) Draw a line of best fit. (1)

(d) Use your line of best fit to estimate the ice-cream sales on a day when the temperature is 25 deg C. (1)

(Total for Question 17 is 5 marks)

18.

A solid cylinder has a radius of 4 cm and a height of 10 cm.

[Diagram: solid cylinder of radius 4 cm and height 10 cm.]

(a) Work out the volume. Give your answer to 1 decimal place. Use pi = 3.142. (3)

(b) Work out the total surface area. Give your answer to 1 decimal place. (2)

(Total for Question 18 is 5 marks)

19.

In a class survey, 90 students were asked to choose their favourite holiday destination.

Destination	Frequency
Spain	30
France	18
Italy	15
USA	12
Australia	9
Other	6

Draw an accurate pie chart to represent this information. Label each sector clearly.

(Total for Question 19 is 5 marks)

20.

The diagram shows an isosceles triangle. The two equal sides each have length (2x + 1) cm. The third side has length (x + 8) cm.

[Diagram: isosceles triangle with two equal sides of length (2x + 1) cm and a third side of length (x + 8) cm.]

The perimeter of the triangle is 49 cm.

(a) Show that 5x + 10 = 49. (2)

(b) Solve to find x. (2)

(c) Hence write down the lengths of the three sides of the triangle. (1)

(Total for Question 20 is 5 marks)

21.

A car cost £18 000 when new.

In the first year, its value falls by 20%.

In each year after that, its value falls by a further 12% per year.

(a) Work out the value of the car at the end of year 1. (2)

(b) Work out the value of the car at the end of year 3. Give your answer to the nearest pound. (4)

(Total for Question 21 is 6 marks)

22.

A fair six-sided dice is rolled twice.

(a) Write down the probability of rolling a 6 on a single roll. (1)

(b) Complete a sample-space diagram or list all 36 outcomes of the two rolls. (No need to write them all - show your method.) (1)

(c) Work out the probability that the total score on the two rolls is exactly 7. (3)

(Total for Question 22 is 5 marks)

TOTAL FOR PAPER: 80 MARKS

Mark Scheme

Foundation Tier - Paper 2 (Calculator) - Set 4 | 80 marks

Question 1

- 45 (%) - B1

Total: 1 marks

Question 2

- 640 x 0.125 or 640 x 12.5 / 100 - M1
- £80 - A1

Total: 2 marks

Question 3

- (a) 5 - B1
- (b) 0.072 - B1

Total: 2 marks

Question 4

- Multiples of 6 or 8 listed (or LCM by prime factorisation) - M1
- 24 - A1

Total: 2 marks

Question 5

- (a) Mode = 1 - B1
- (b) Sum(fx) = 0 + 7 + 12 + 9 + 4 - M1
- (b) 32 goals - A1

Total: 3 marks

Question 6

- Multiplying first term: 3 x 2x = 6x or 3 x (-5) = -15 - M1
- 6x - 15 - A1

Total: 2 marks

Question 7

- 1 hr 20 min = 80 min or 1.333... hr - M1
- 30 / (4/3) or 30 / 80 x 60 - M1
- 22.5 (km/h) - A1

Total: 3 marks

Question 8

- (a) Middle two values 30 and 33 identified - M1
- (a) (30 + 33) / 2 = 31.5 - A1
- (b) 52 - 18 = 34 - B1

Total: 3 marks

Question 9

- 288 / (5 + 3 + 4) = 24 - M1
- 3 x 24 - M1
- Mia = £72 - A1

Total: 3 marks

Question 10

- Variable terms collected (e.g. x = 13) - M1
- Linear simplification clear - M1
- x = 13 - A1

Total: 3 marks

Question 11

- Area = (1/2) x (a + b) x h written - M1
- (1/2) x (7 + 11) x 6 - M1
- 54 (cm^2) - A1

Total: 3 marks

Question 12

- (a) 144 / 360 x 180 - M1
- (a) 72 (people) - A1
- (b) 72 / 360 seen - M1
- (b) 1/5 - A1

Total: 4 marks

Question 13

- (a) 3 x 4 = 12 seen - M1
- (a) v = 8 + 12 = 20 - A1
- (b) 86 - 32 = 54 or 5 x 54 / 9 - M1
- (b) C = 30 - A1

Total: 4 marks

Question 14

- (a) Increase = 5085 - 4500 = 585 - M1
- (a) 585 / 4500 x 100 - M1
- (a) 13.0 (%) - A1
- (b) Multiplier 1.08 seen - M1
- (b) 3800 x 1.08 = 4104 - A1

Total: 5 marks

Question 15

- (a) 23 - B1
- (b) Common difference 4 noted (4n + ...) - M1
- (b) 4n + 3 - A1
- (c) 4 x 50 + 3 - M1
- (c) 203 - A1

Total: 5 marks

Question 16

- (a) 600 / 8 x 14 or 75 x 14 - M1
- (a) 1050 (g) - A1
- (b) 1050 / 75 attempted - M1
- (b) 14 jars - A1

Total: 4 marks

Question 17

- (a) All 8 points plotted correctly - B2 (B1 if 6+ correct)
- (b) Positive correlation (strong / very strong) - B1
- (c) Sensible line of best fit - B1
- (d) Sales estimated at 25 deg C (accept £270 - £290) - B1

Total: 5 marks

Question 18

- (a) pi x 4^2 x 10 or 3.142 x 16 x 10 - M1
- (a) 502.72 evaluated - M1
- (a) 502.7 (cm^3) accept 502.4 - 502.9 - A1
- (b) 2 x pi x 4^2 + 2 x pi x 4 x 10 (= 100.544 + 251.36) - M1
- (b) 351.9 (cm^2) accept 351.7 - 352.1 - A1

Total: 5 marks

Question 19

- 360 / 90 = 4 deg per student - M1
- At least 3 angles correct: Spain 120, France 72, Italy 60, USA 48, Australia 36, Other 24 (deg) - M1
- All 6 angles correct - A1
- Pie chart drawn within 2 deg tolerance - B1
- All sectors labelled - B1

Total: 5 marks

Question 20

- (a) Perimeter = 2(2x + 1) + (x + 8) or 4x + 2 + x + 8 - M1
- (a) 5x + 10 = 49 derived - C1
- (b) 5x = 39 - M1
- (b) x = 7.8 - A1
- (c) Two sides 16.6 cm and third 15.8 cm - B1

Total: 5 marks

Question 21

- (a) Multiplier 0.8 seen - M1
- (a) 18000 x 0.8 = £14400 - A1
- (b) Year 2 multiplier 0.88 seen - M1
- (b) 14400 x 0.88 = £12672 - M1

- (b) 12672 x 0.88 - M1
- (b) £11151 - A1

Total: 6 marks

Question 22

- (a) 1/6 - B1
- (b) Sample space size 6 x 6 = 36 stated or grid started - B1
- (c) Pairs giving total 7: (1,6),(2,5),(3,4),(4,3),(5,2),(6,1) - six pairs - M1
- (c) 6/36 written - M1
- (c) 1/6 - A1

Total: 5 marks

Worked Solutions

Foundation Tier - Paper 2 (Calculator) - Set 4 | Detailed explanations

Question 1

To convert a decimal to a percentage, multiply by 100.
0.45 x 100 = 45%.

Question 2

12.5% = 12.5 / 100 = 0.125.
0.125 x 640 = 80.
Answer: £80.
Mental check: 12.5% is 1/8 of the amount. 640 / 8 = 80.

Question 3

(a) 1 significant figure of 4.836: the first non-zero digit is 4. Next digit is 8 (>= 5) so round up: 5.
(b) 2 significant figures of 0.0719: leading zeros do not count. First two sig figs are 7 and 1. Next digit is 9 (>= 5), round up to 0.072.

Question 4

List multiples of 6: 6, 12, 18, 24, 30, 36, ...
List multiples of 8: 8, 16, 24, 32, 40, ...
First common multiple is 24.
LCM = 24.

Question 5

(a) The mode is the value with the highest frequency. Frequency 7 occurs at goals = 1, so mode = 1.
(b) Total goals = Sum(fx) = 0x3 + 1x7 + 2x6 + 3x3 + 4x1 = 0 + 7 + 12 + 9 + 4 = 32.

Question 6

Multiply each term inside the bracket by 3.
3 x 2x = 6x. 3 x (-5) = -15.
Answer: 6x - 15.

Question 7

Convert time to hours: 1 hr 20 min = 1 + 20/60 = 4/3 hours (approx 1.333 hr).
Speed = distance / time = 30 / (4/3) = 30 x 3/4 = 22.5 km/h.

Question 8

(a) 10 values, so median is the average of the 5th and 6th: (30 + 33) / 2 = 31.5.
(b) Range = largest - smallest = 52 - 18 = 34.

Question 9

Total parts = 5 + 3 + 4 = 12.
One part = £288 / 12 = £24.
Mia gets 3 parts = 3 x 24 = £72.

Question 10

3x + 4 = 2x + 17.
Subtract 2x: x + 4 = 17.
Subtract 4: x = 13.
Check: 3(13) + 4 = 43 and 2(13) + 17 = 43.

Question 11

Area of trapezium = (1/2) x (a + b) x h, where a and b are the parallel sides and h the perpendicular height.
Area = (1/2) x (7 + 11) x 6 = (1/2) x 18 x 6 = 9 x 6 = 54 cm^2.

Question 12

(a) Train: (144 / 360) x 180 = 0.4 x 180 = 72 people.
(b) Cycle: 72 / 360 = 1/5.

Question 13

(a) v = u + at = 8 + 3 x 4 = 8 + 12 = 20.
(b) C = 5(F - 32) / 9 = 5(86 - 32) / 9 = 5 x 54 / 9 = 270 / 9 = 30.
So C = 30 deg.

Question 14

(a) Increase = 5085 - 4500 = 585.
% increase = (585 / 4500) x 100 = 13.0%.
(b) 8% increase on 3800: multiplier 1.08, so 3800 x 1.08 = 4104.

Question 15

(a) Common difference is +4. Next term = 19 + 4 = 23.
(b) Zeroth term = 7 - 4 = 3. So nth term = 4n + 3.
(c) 50th term = 4 x 50 + 3 = 200 + 3 = 203.

Question 16

(a) Per portion, tomatoes = 600 / 8 = 75 g. For 14 portions: 75 x 14 = 1050 g.
(b) Each jar replaces 75 g of tomatoes. Jars needed = 1050 / 75 = 14 jars.

Question 17

(a) Plot 8 points.
(b) Positive correlation - higher temperature, higher sales.
(c) Best fit drawn through the data points.
(d) At 25 deg C the line gives roughly £270 - £290.

Question 18

(a) Volume = pi x r^2 x h = 3.142 x 16 x 10 = 502.72.
To 1 dp: 502.7 cm^3.
(b) Surface area = 2 x pi x r^2 + 2 x pi x r x h (two circular ends + curved side).
= 2 x 3.142 x 16 + 2 x 3.142 x 4 x 10 = 100.544 + 251.36 = 351.904.
To 1 dp: 351.9 cm^2.

Question 19

Angle per student = 360 / 90 = 4 deg.
Spain: 30 x 4 = 120 deg.
France: 18 x 4 = 72 deg.
Italy: 15 x 4 = 60 deg.
USA: 12 x 4 = 48 deg.
Australia: 9 x 4 = 36 deg.
Other: 6 x 4 = 24 deg.
Check: 120 + 72 + 60 + 48 + 36 + 24 = 360 deg.

Question 20

(a) Two equal sides + third side = perimeter.
Perimeter = 2(2x + 1) + (x + 8) = 4x + 2 + x + 8 = 5x + 10.
Setting equal to 49: 5x + 10 = 49. (Shown.)
(b) 5x = 39, so x = 7.8.
(c) Equal sides: 2(7.8) + 1 = 16.6 cm. Third side: 7.8 + 8 = 15.8 cm.
Check: 16.6 + 16.6 + 15.8 = 49 cm.

Question 21

(a) Year 1: 20% loss, multiplier 0.8.
Value after year 1 = 18000 x 0.8 = £14400.
(b) Years 2, 3: each 12% loss, multiplier 0.88.
After year 2: 14400 x 0.88 = £12672.
After year 3: 12672 x 0.88 = £11151.36.

To the nearest pound: £11151.

Question 22

(a) P(6) = 1/6.

(b) The two rolls give 6 x 6 = 36 equally likely ordered outcomes.

(c) Pairs that sum to 7: (1,6), (2,5), (3,4), (4,3), (5,2), (6,1). That is 6 pairs.

P(total = 7) = 6 / 36 = 1/6.